Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer with 2+ years building **cloud-native full-stack services**, **distributed data workflows**, and **reliable CI/CD automation** using Python, React, SQL, Docker, and observability-driven debugging practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **scalable data services and user interfaces** for 500+ users, reducing query latency from **1.2s to 350ms** through indexing, API optimization, and runtime profiling.
- Implemented **fault-tolerant orchestration** for 5K+ monthly jobs with validation and rollback safeguards, improving reliability and incident response across distributed pipelines.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **React + Flask production features** handling 10K+ daily requests, translating product requirements into maintainable APIs/components with secure, tested release behavior.
- Improved **CI/CD quality gates** and defect-triage workflows, reducing turnaround by **60%** while sustaining availability and customer-facing functionality under continuous delivery.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **large-scale experimentation pipelines** over 100K+ trajectories, enabling reproducible evaluations and structured diagnostics for data-intensive model behavior analysis.
- Created **technical reports and observability summaries** to bridge research and engineering teams, accelerating issue isolation and improving implementation planning.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **distributed API and ETL services** with Python/Java, AWS Glue, and S3, improving response time by **30%** and strengthening cross-system integration reliability.
- Maintained **20+ automated CI test suites** using Jenkins, PyTest, and Mockito, boosting pipeline efficiency by **68%** and reducing regression defects across releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented **MERN + WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving consistency and responsiveness in distributed user operations.
- Automated **containerized deployment workflows** with Docker/Jenkins, reducing deployment time by **50%** and improving release repeatability in staging and production.

Skills

Languages: **Python**, Java, JavaScript/TypeScript, SQL, C++, Bash
Frontend: **React.js**, HTML5, CSS3, accessibility-aware component design
Backend: **Flask**, **FastAPI**, REST APIs, distributed services, PostgreSQL
Cloud & Ops: AWS (S3, Glue, RDS), Docker, Kubernetes basics, Jenkins, GitHub Actions, CI/CD
Engineering: testing automation, on-call readiness mindset, troubleshooting, documentation, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **high-throughput fintech platform** supporting 50K+ users and 1M+ transactions with resilient backend services, secure integrations, and scalable cloud deployment workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed a **schema-aware query assistant** across 10+ databases with validation checks and structured outputs, improving productivity for data retrieval and debugging use cases.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized processing service** on AWS ECS/Fargate with automated CI/CD validation, improving production reliability and operational continuity for data pipelines.